UpWork Requirement Sheet

10/19/24

Hey!

I thought we had a great conversation today. I will use this document to go over the requirements, deliverables, some resources, and photos. Because of all the information in here, I'm anticipating this build will be fairly straightforward, and would like to budget 10-13 hours for this.

Because the final diffuser model is subject to change, I think we can hold off on doing a complete PCB design in this initial process. We could do a preliminary PCB design, but no need for the deliverable to be a completed PCB design as I would like to submit finish geometry to you before we do that.

Deliverables

- 2 physical, functioning circuits (could be printed or manual, whatever is easier for you)
  - If you have the ability to do the PCB route and print just two of them as a test run, this would definitely be preferred (as that is how the final circuit will be made).
  - includes the circuit board itself along with the component parts
- a preliminary PCB design with parts and connections mapped

Requirements

CIRCUIT GEOMETRY A

- akin to IMAGEA, a circle with a maximum diameter of 5" with space on the inside for the atomizer disc

CIRCUIT GEOMETRY B

- a circle with max diameter of 4"

- both circuits should have 3 small holes for attachment to the body
- 24V input jack
- entire circuit powered by a binary switch
- on the ON → the following
  - a small DC fan
  - an ultrasonic atomizer and associated circuitry

Extra Credit

- Most essential oil diffusers have some type of stop mechanism either as a max amount of time, or a water sensor to only function if there is water in the containment tank. I'd like you to do a bit of research (~1hr) on the different feasible options and schedule some time to discuss. This isn't a mandatory feature but I would like to have.
- some type of audio / beeper that does a short beep on the "ON" and a longer beep on the "OFF".

Asks

- Please take photos, videos, and screenshots along the way so that I can see your progress and process. This is for me to feel good about compensating you, and for my own curiosity.
- Please keep receipts of any parts and materials you use for reimbursement.

Resources

PARTS

I would like to find these parts on whatever site or vendor makes sourcing these easy. But for now these are easy to source. No need to use these exact ones if you find better ones.

https://www.amazon.com/MEETOOT-2-Pack-Ultrasonic-Ceramics-Sealing/dp/B09L7KBF53/ref=asc_df_B09L7KBF53/?tag=hyprod-20&linkCode=df0&hvadid=693129581498&hvpos=&hvnetw=g&hvrand=9099652171037963 43&hvpone=&hvptwo=&hvqmt=1792948580792&mcid=5b04078bac4d3728a1a35735f5d597d0&th=1

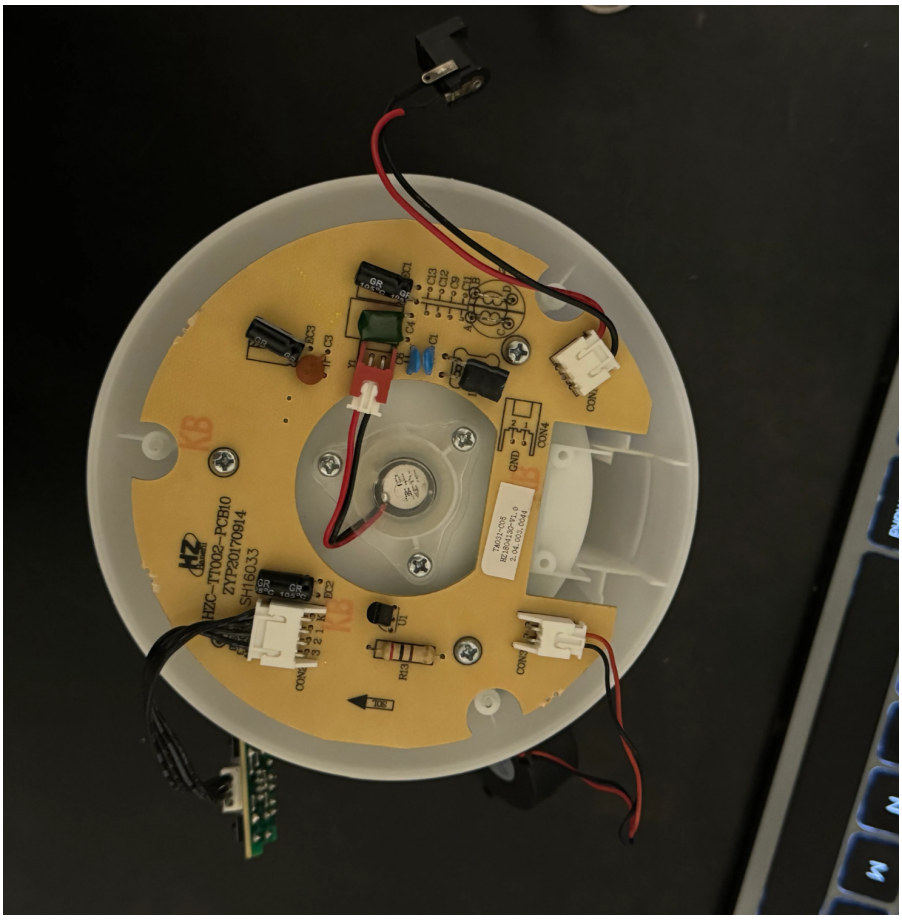
https://www.amazon.com/Atomization-Humidifier-Driver-Ultrasonic-Circuit/dp/B0BX6CV5P3/ref=asc_df_B0BX6CV5P3/?tag=hyprod-20&linkCode=df0&hvadid=692875362841&hvpos=&hvnetw=g&hvrand=9099652171037963 43&hvpone=&hvptwo=&hvqmt=2281435178138&psc=1&mcid=2a2b04ec653e3fb2a178ad3d3accd2f6&hvocijid=909965217103796343-B0BX6CV5P3-&hvexpln=73

^ this second option has the associated oscillator circuit, for your reference.

https://www.amazon.com/Gardner-Bender-GSW-18-Heavy-Duty-Electrical/dp/B00004WLKC/ref=asc_df_B00004WLKC/?tag=hyprod-20&linkCode=df0&hvadid=692875362841&hvpos=&hvnetw=g&hvrand=16593478342789992776&hvpone=&hvptwo=&hvqmt=2281435182458&mcid=781942f011a53cb6a4fb930694abdc59&hvocijid=16593478342789992776-B00004WLKC-&hvexpln=73&th=1

https://www.amazon.com/Wathai-Brushless-Cooling-Turbo-Blower/dp/B07RP14KML?source=ps-sl-shoppingads-lpcontext&ref_=fplfs&smid=A1AV40UDU1XL8S&th=1

IMAGE A



- See how this diffuser circuit has connectors and attachments between the parts and the PCB, to make assembly easier. I think this would be a bonus and would make sense, but isn’t mandatory.
- Note that this circuit has LEDs mounted on the opposite side. This is not necessary for this iteration of the circuit (although I am interested in the future to have a few surface mounted LEDs at 2700K color temp).

IMAGEB



- I believe the microprocessors here are for handling a touch capacitive and multi option color settings.